Online Compiler Website Requirements

Project Title: Web-Based Online Compiler and Code Execution Platform

1. Project Overview

An online compiler allows users to write, compile, and execute code directly from a web browser without installing software.

2. Functional Requirements

User Registration, Login/Logout, Profile Management, Password Reset.

Code Editor: Syntax Highlighting, Line Numbers, Auto Indentation, Auto Save, Theme Selection.

Compiler Module: C, C++, Java, Python, JavaScript support, Compile, Execute, Output and Error Display.

File Management: Create, Save, Open, Delete, Download Programs.

Admin Module: Manage Users, Languages, Activity Logs.

3. Non-Functional Requirements

Fast Response Time, Secure Execution, Responsive Design, Data Backup, User-Friendly Interface.

4. Software Requirements

Frontend: HTML, CSS, JavaScript, Bootstrap. Backend: PHP/Java/Python/Node.js. Database: MySQL. Editor: Monaco or Ace.

5. Hardware Requirements

Server: Intel i5+, 8GB RAM, 100GB Storage. Client: Browser-enabled Computer or Mobile.

6. Database Tables

Users, Programs, Execution_History.

7. Additional Features

AI Code Suggestions, Coding Challenges, Online Tests, Certificate Generation, Code Sharing.

8. Project Modules

Authentication, Code Editor, Compiler Integration, Output Display, File Management, Admin Panel, Analytics.